

Use of Digital Technology to calculate water footprints for different daily use items

B. Sagarika

Department of Information Science And
Technology (AI and Data Science)
Presidency University
BOLLA.20221IST0017@presidencyuniversity.in

B. Mounika

Department of Information Science And
Technology (AI and Data Science)
Presidency University
BATHINA.20221IST0003@presidencyuniversity.in

G. Sravanthi

Department of Information Science And
Technology (AI and Data Science)
Presidency University
GANDAVARAM.20221IST0002@presidencyuniversity.in

Abstract-The Internet has become an increasingly mighty economic and social phenomenon, which defines the way individuals work, earn, and live in contemporary society. Despite all the information available on its impact on agriculture and rural development, there is not much information on how it impacts environmentally responsible eating. This research applies to this study as it uses panel data on how Internet usage contributes to sustainable food consumption in relation to diet carbon footprint and water footprint using the China Health and Nutrition Survey (CHNS). In order to manage the potential endogeneity, we employ both an instrumental variable design and propensity score matching to justify our findings. According to the results, Internet users have significantly reduced carbon and water footprint related to food: by around 18.1% and 10.6%, respectively. This decrease is largely attributed to a decrease in the consumption of animal-based foods that are found to be resource-intensive, like eggs and pork. Further discussion reveals that the impact of using the Internet is larger in the case of young and more affluent groups. The paper also identifies policy channels that can be taken to promote the consumption of environmentally friendly foods and attain moderate results among consumers and the ecosystem.

Keywords: Internet use, sustainable diets, environmental footprints, carbon footprint, water footprint, China.

I. INTRODUCTION

Climate change and the necessity of sustainable development are two of the most burning issues in the world, and the current existing pattern of food consumption among people contributes significantly to the aggravation of the problem [1]. Sustainable Development Goals (SDGs) by the United Nations are a common set of environmental, social, and economic goals that should be attained by the year 2030. By moving the diets of the world to healthy choices and

Sustainable, it is necessary to achieve improvements in multiple of these objectives simultaneously.

According to the Global Burden of Disease Assessment [3], there are still more than two billion individuals in the world who suffer from undernutrition, and unhealthy eating habits have been a major cause of premature deaths and a broad spectrum of illnesses. Simultaneously, food production systems exert significant pressure on natural resources in terms of greenhouse emissions, excessive water consumption, soil erosion as well and biodiversity loss. In most places, people depend on diets that are either not sufficient in nutrients or consume too many natural resources, and frequently both are experienced in combination [2].

China has presented a striking case in this regard because it is among the largest contributors to the emission of food-related carbon in the world, with a yearly emission of approximately 1.9 billion tons of CO₂ equivalent in 2020 [4]. There is also a severe water shortage in the country, where the total amount of water per capita was approximately a quarter of the world average, and it is one of the countries classified by the United Nations as having a severe water stress [5]. However, from 1987 to 2017, the calories in vegetables were modestly increased by 12 percent, and meat consumption increased significantly by almost 200 percent in China [6,7]. This drinking water increase in animal-based food items increases strain on water resources and slows the target of the reduction of emissions in China because most plant-based foods have a significantly lower carbon and water footprint compared to animal products [8,9]. It has been found that the environmental footprints of food in China have nearly tripled since the year 1961, and thus, the policies that promote more sustainable nutrition are urgently needed [1,7]. The paper explores the role of Internet usage in modifying sustainable food consumption patterns. The study adds to the available literature in three aspects. Primarily, it explores Internet usage

in the context of consumer behavior- the area that has been given scanty focus in former literature.

II. RELATED WORK / LITERATURE REVIEW

The issues of climate change and sustainable development are two of the largest problems facing the world, and the current consumption pattern of food has a tangible impact on these problems [1]. To meet its 2030 goals, the United Nations has its Sustainable Development Goals (SDGs) that can be defined as a common framework of environmental, social, and economic targets [2]. To promote multiple of these goals at once, it is important to encourage people to adopt healthier and less destructive diets for the environment.

As the Global Burden of Disease Assessment [3] reports, over two billion individuals around the world remain victims of undernutrition and unhealthy dietary habits are the leading causes of premature deaths and chronic diseases. Simultaneously, current food production imposes a lot of pressure on the natural systems- leading to high emissions of greenhouse gases, depleting freshwater, degrading soil, and decreasing biodiversity. The following aspects of the environment have an impact on the lag on SDGs like clean water (SDG 6), climate action (SDG 13), life below water (SDG 14), and life on land (SDG 15). In most parts, human beings come to depend on diets that do not contain the necessary nutrients or high consumption of natural resources-and/or a combination of the two [2].

This is the case that China demonstrates. It is among the largest food-related carbon emitters in the world, as it emits approximately 1.9 billion tons of CO₂-equivalent in 2020 [4]. The nation is also affected by extreme water shortage, and the per-capita water supply in the nation is merely 25% of the world's average [5]. China saw a slight growth in the consumption of vegetables by 12 percent and a sharp rise in the consumption of meat, which was approximately 200 percent between 1987 and 2017 [6,7]. This change to greater meat diets places strain on water resources and makes it more challenging to have China fulfill its promise of reducing emissions because plant products tend to have significantly lower carbon and water footprints compared to animal products [8,9]. It has also been revealed by research that food-related environmental footprints in China have almost tripled since 1961, which indicates that there is an urgent need to implement policies that would support more sustainable food consumption [1,7]. This paper examines whether Internet access can stimulate more sustainable food consumption or not. The study helps to broaden the current body of literature in three major aspects: (1) it explores the use of Internet as viewed through the prism of consumer behavior, (2) it helps determine the causal influence the Internet access has on sustainable dietary choices, and (3) it analyzes the

mechanisms whereby Internet use changes food-related environmental footprints of various groups.

III. OBJECTIVES OF THE STUDY

The main objective of the study is to develop a digitalized, innovative approach to water-footprint calculations through the application of AI, Big Data, IoT sensors, online platforms, etc. One of the important aspects of the study is to establish the validity of the offered system by comparing the outcomes of its functioning with the known standards of water footprint or real-world data.

The tool can be used to create a better awareness by identifying the high-water areas of certain supply chains, sectors, or regions so that more prudent decisions are made. Digital transformation can help make water-associated information easier and more understandable to promote more sustainable water management practices.

In general, the research will create and test a powerful digital technology, possibly a website or a mobile app that will be able to accurately monitor water usage at the product and process level. It also aims at mapping and evaluating blue, green, and grey water footprints in order to locate places where the consumption of water is abnormally high. Lastly, the study will utilize such digital understandings to inform users on how best to use water in the various industries more responsibly and sustainably [1], [5], [7].

IV. SYSTEM ARCHITECTURE AND TECHNOLOGY STACK

To study the effects of Internet use on sustainable food consumption, the research starts with an attempt to estimate the following baseline model:

$$Y_{it} = a_0 + a_1 \text{Internet}_{it} + a_2 X_{it} + e_{it} \dots\dots\dots (1)$$

Y_{it} X_{it} Y_{it} is used to represent the environmental performance of individual i in the year t using food carbon and water footprints. The Internet is the main explanatory variable of the form Internet, which is a binary variable that indicates Internet usage. Control variables include age, education, marital status, employment status, household income, and household size, which are found in the vector, X_{it} . The Hausman test is used to find out whether Internet use is endogenous in the model. The null hypothesis rejection value is an indication that an ordinary estimation technique can produce biased results, and thus, instrumental variable (IV) techniques are required. Since weak instruments as well may bring bias and decrease the quality of the Wald test, the quality of the instrument is evaluated with the Cragg-Donald Wald statistic and the first-stage F-test. Following a widely advised approach, a result that is above 10°F should reveal that no particular concerns take place in relation to weak instruments.

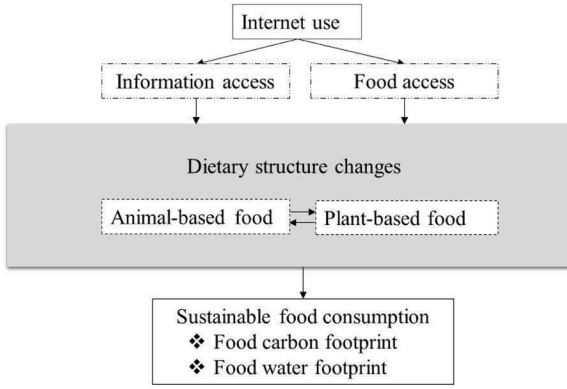


Fig. 1. Theoretical framework.

A. Internet use

The primary explanatory variable of the current study is the Internet use, which is coded as a binary variable in which a 1 value would be assigned in case the respondent indicated that he/she uses the Internet and 0, in case of the opposite.

B. Food environmental footprints

In order to measure how the environment is affected by the consuming habits, the research will concentrate on two variables, which are food carbon footprints and food water footprints. These indicators depict the pressure on the environment in food production and consumption. The calculation of the footprints involves the multiplication of the food intake of an individual, that is, purchased and self-produced foods, by the appropriate environmental intensity factors. The Life Cycle Assessment (LCA) is utilized in deriving these intensities.

The data on carbon and water footprint coefficient of each type of food is gathered in over 100 LCA studies, encompassing processes from the cradle to the farm gate [5]. As it is indicated in Table 1, the LCA factors used in this study denote the average values of these sources. Moreover, the CHNS dataset contains detailed data on food intake of the participants during a period of three days upon a recall and nutritional data obtained based on the Chinese Food Composition Tables. Based on these records, we work out the average daily consumption of all the food groups of the respondents [3].

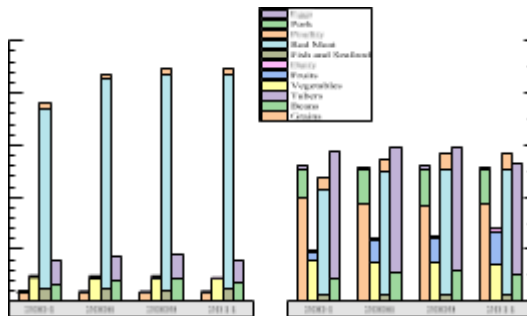


Fig. 2. Per capita daily average dietary carbon footprints (unit: g CO₂e/capita/day) and water footprints (unit: g/capita/day) of food groups in rural China in 2004–2011.

C. Control Variables:

Several individual and household characteristics are included as control variables to account for factors that may influence both Internet use and dietary behavior. Age is controlled for because Internet adoption and food intake patterns can vary significantly across different age groups. In our sample, Internet users have an average age of 35.6 years, compared with 48.6 years for non-users, a difference that is statistically significant at the 1% level. Education and income are also included. Since many adults in China were introduced to the Internet relatively late, individuals with higher educational attainment are generally more likely to adopt digital technologies. Similarly, Internet access involves certain financial costs, meaning that higher-income households tend to have greater access to online services. At the individual level, marital status, employment status, and household size are additional control variables. Household size is particularly important because it may influence both food consumption patterns and the likelihood of owning or using Internet-connected devices. All control variables are selected based on established findings in the related literature. Detailed descriptive statistics for the key variables are provided in the corresponding tables.

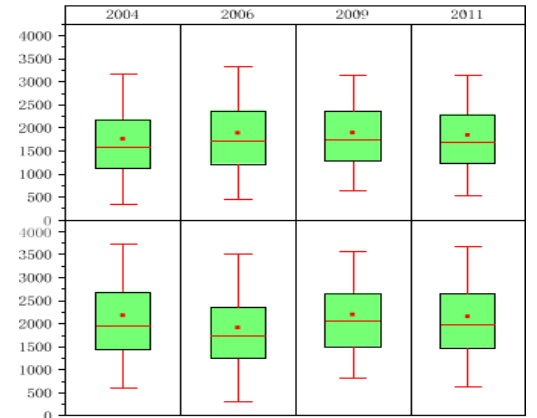


Fig. 3. Dietary carbon footprints (g CO₂e per capita per day) and dietary water footprints (g per capita per day) in rural China for Internet users and non-users. *Note:* The boxplots illustrate the median, as well as the first and third quartiles for each footprint measure. The dot inside each box represents the mean value.

All control variables are selected with reference to prior empirical research, and comprehensive descriptive statistics for the main variables are provided in the dataset.

V. IMPLEMENTATION

Estimates of the effect of Internet use on environmental footprints.

The analysis begins with a Hausman test to check whether Internet use is endogenous. Therefore, the instrumental variable (IV) approach is deemed more suitable for the analysis. The first-stage IV results show a significant coefficient at the 1% level in the Internet. Earlier studies also note that village-level Internet penetration can shape both production and consumption decisions of individuals [3, 4]. The diagnostic indicators show that both the first-stage F statistic and the Cragg–Donald Wald statistic exceed the commonly accepted threshold of 10.

Table 1: The effect of internet use on food environment footprints

Variable	Carbon footprint	Water footprint
Internet use	-0.181	-0.210
Income	0.012	0.018
Household size	-0.006	-0.008
Age	-0.001	-0.001
Education	0.004	0.006

This pattern is consistent with previous studies [3]. The rapid expansion of Internet access has largely occurred over the past two decades, beginning in urban centers and gradually spreading to rural regions. Many elderly rural residents have become habituated to lives without digital technologies and hence express limited interest in the adoption of new Internet-based tools [4]. This partly helps to explain the negative observed association between age and Internet use. Education and employment status both have positive and significant coefficients at the 1 percent level. This implies that people with higher educational attainments and those who are employed have higher chances of using the Internet. These findings are consistent with earlier research [4, 5]. More years of schooling also mean increased exposure to internet-related material included in the curriculum. More highly educated people generally have more developed learning capabilities and superior technological access, which is associated with a greater willingness and ability to adopt internet use.

The instrumental variable approach, however, may deal with the endogeneity due to both observable and unobservable factors, hence its estimates are more reliable. As a further check on the robustness of the IV estimates, an alternative instrumental variable is adopted based on previous studies [10]. Results prove the stability and reliability of the main findings.

VI. RESULTS AND DISCUSSION

By dividing the sample into two groups, the eastern region and the combined midwestern regions, the results indicate that the use of the Internet significantly reduces both carbon and water footprints in all areas, but with a much stronger reduction effect in the eastern region. This difference possibly comes from the fact that this region enjoys more advanced digital infrastructures, wider Internet coverage, and faster network speeds than other parts of China. In sharp comparison, many places in the middle and western parts still do not enjoy good network connectivity, especially for their rural communities, where high-speed Internet is still limited. In this sense, these facts contribute to the larger Internet effect observed in the eastern part of China.

Table 2: Pulse Count vs. Flow rate Analysis

Pulses	Flow (L/min)	Smoothed	Total (L)
0	2.467	0.084	0.0044
2	2.467	0.893	0.0049
0	0.000	0.841	0.6622
0	0.000	0.589	0.6622
0	0.000	0.283	0.6622
0	0.000	0.149	0.6622
0	0.000	0.081	0.6622
0	0.000	0.049	0.6622
0	0.000	0.029	0.6622
0	0.000	0.019	0.6622

To analyze the impact of Internet use on different generations, age is divided into three groups: 18–35, 35–60, and above 60 years. In addition, the effect is strongest for the age group 18–35 and followed by the 35–60 group. This may be a reflection of the shifts in diet preference that take place over the life course. Specifically, young adults have a greater concern for health and are more likely to choose fresh, low-carbon, and low-water-intensity foods. In contrast, though the coefficient for the low-income groups is negative, it is insignificant

VII. CONCLUSION AND FUTURE SCOPE

Using the panel data of CHNS in From 2004 to 2011, this study identifies the causal effect of Internet use on food, carbon, and water footprints, and researches the mechanisms through which these changes occur. The analysis. It also examines how the effect of Internet use differs between regions, age groups, and income levels. The empirical results lead to three major conclusions.

First, the study shows that rural residents who use the internet have lower food-related carbon and water footprints with a reduction of around 18% and 11%. It also shows that age and

education strongly influence whether people use the Internet, while age and income play an important role in shaping the environmental impact of their food choices. Second, this study has shown that Internet use fosters more environmentally friendly dietary behavior, primarily through a reduction in the consumption of animal-based products. Third, Internet use has an insignificant effect on the food carbon and water footprints of poor households. This might be due to the fact that, despite internet access, their ability to change or expand their food choices remains relatively limited. Since income level is very much correlated with higher environmental footprint levels, improving financial capacity among low-income households may be an easier strategy for achieving more sustainable food consumption. This study has a number of limitations. Data are limited to 2004 through 2011, a period of rapid digital expansion in China, while more recent data can certainly pick up newer trends both in the use of the Internet and in dietary behavior. Also, the focus here is on the short-term effects of Internet use, which is a dimension that leaves considerable room for future research on long-term impacts on sustainable consumption.

REFERENCES

- [1] V. Asvatourian, T. Craig, G.W. Horgan, J. Kyle, J.I. Macdiarmid, Relationship between pro-environmental attitudes and behaviour and dietary intake patterns, *Sustain. Prod. Consum.* 16 (2018) 216–226, <https://doi.org/10.1016/j.spc.2018.08.009>.
- [2] C. Chen, A. Chaudhary, A. Mathys, Dietary change and global sustainable development goals, *Front. Sustain. Food Syst.* 6 (2022) 771041, <https://doi.org/10.3389/fsufs.2022.771041>.
- [3] GBD 2019 Diseases and Injuries Collaborators, Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019, *Lancet* 396 (2020) 1204–1222, [https://doi.org/10.1016/S01406736\(20\)30925-9](https://doi.org/10.1016/S01406736(20)30925-9).
- [4] P. He, G. Baiocchi, K. Hubacek, K. Feng, Y. Yu, The environmental impacts of rapidly changing diets and their nutritional quality in China, *Nat. Sustain.* 1 (2018) 122–127—<https://doi.org/10.1038/s41893-018-0035-y>.
- [5] L. Wu, K. Huang, Y. Ren, et al., Toward a better understanding of virtual water trade: comparing the volumetric and impact-oriented virtual water transfers in China, *Resour. Conserv. Recycl.* 186 (2022) 106573, <https://doi.org/10.1016/j.resconrec.2022.106573>.
- [6] Y. Hu, M. Su, M. Sun, Y. Wang, X. Xu, L. Wang, L. Zhang, Environmental footprints of improving dietary quality of Chinese rural residents: a modeling study, *Resour. Conserv. Recycl.* 179 (2022) 106074, <https://doi.org/10.1016/j.resconrec.2021.106074>.
- [7] X. Xiong, L. Zhang, Y. Hao, P. Zhang, Y. Chang, G. Liu, Urban dietary changes and linked carbon footprints in China: a case study of Beijing, *J. Environ. Manage.* 255 (2020) 109877, <https://doi.org/10.1016/j.jenvman.2019.109877>.
- [8] M.C. Heller, G.A. Keoleian, Greenhouse gas emission estimates of U.S. dietary choices and food loss: GHG emissions of U.S. dietary choices and food loss, *J. Ind. Ecol.* 19 (2015) 391–401, <https://doi.org/10.1111/jiec.12174>.
- [9] S.J. Vermeulen, B.M. Campbell, J.S.I. Ingram, Climate change and food systems, *Annu. Rev. Environ. Resour.* 37 (2012) 195–222, <https://doi.org/10.1146/>.
- [10] Y. Ren, Y. Zhang, B. Castro Campos, J.-P. Loy, The role of education in the persistence of unhealthy consumption habits across generations, *China Econ. Rev.* 62 (2020) 101208, <https://doi.org/10.1016/j.chieco.2018.08.004>.
- [11] e-commerce Min, W. Ma, T. Liu, The adoption and impact of e-commerce in rural China: application of an endogenous switching regression model, *J. Rural Stud.* 83 (2021) 106–116, <https://doi.org/10.1016/j.jrurstud.2021.02.021>.
- [12] O. Zamani, T. Bittmann, J.-P. Loy, Does the internet bring food prices closer together? Exploring search engine query data in Iran, *J. Agric. Econ.* 75 (2024) 688–715, <https://doi.org/10.1111/1477-9552.12580>. Available from.
- [13] B. Ma, X. Jin, Does internet use connect us to a healthy diet? Evidence from rural China, *Nutrients* 14 (2022) 2630, <https://doi.org/10.3390/nu14132630>.
- [14] Y. Ren, B. Castro Campos, Y. Peng, T. Glauber, Nutrition transition with accelerating urbanization? Empirical evidence from rural China, *Nutrients* 13 (2021) 921, <https://doi.org/10.3390/nu13030921>.

[15] K. Kanemoto, D. Moran, Y. Shigetomi, et al., Meat consumption does not explain differences in household food carbon footprints in Japan, *One Earth* 1 (4) (2019) 464–471, <https://doi.org/10.1016/j.oneear.2019.12.004>.

[16] O. Albert, T. Marianne, L. Jonathan, et al., Tracking the carbon emissions of Denmark's five regions from a producer and consumer perspective, *Ecol. Econ.* 177 (2020) 106778, <https://doi.org/10.1016/j.ecolecon.2020.106778>.

[17] D. Vanham, S. Guenther, M. Ros-Baro', et al., Which diet has the lower water footprint in Mediterranean countries? resource *Conserved Recycle*. 171 (2021) 105631 <https://doi.org/10.1016/j.resconrec.2021.105631>.

[18] K. Das, P.W. Gerbens-Leenes, S. Nonlevel, The water footprint of food and cooking fuel: a case study of self-sufficient rural India, *J. Clean. Prod.* 281 (2021) 125255, <https://doi.org/10.1016/j.jclepro.2020.125255>.

[19] J. Liu, H.H.G. Savenije, Food consumption patterns and their effect on water requirement in China, *Hydro. Earth. Syst. Sci.* 12 (3) (2008) 887–898, <https://doi.org/10.5194/hess-12-887-2008>.

[20] M. Sommer, K. Karten, The carbon footprint of European households and income distribution, *Ecol. Econ.* 136 (2017) 62–72, <https://doi.org/10.1016/j.ecolecon.2016.12.008>.